

Real-time process management system in a restaurant by sharing food order information

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Abstract—In a full-service restaurant, it is crucial to share order information among staff in the dining room and kitchen. This paper introduces a real-time process management system for restaurants using an advanced point-of-sale (POS) system by which staff can share order information in real time. In this system, kitchen staff can check all customer orders by the dish that was ordered and the elapsed time of each order. Moreover, dining hall workers can grasp their customer situation with a monitor. By introducing this system to a restaurant, we confirmed that it can make preparation processes more efficient and reduce customers' claims.

Keywords—component; restaurant service; point-of-sale system; process management; information sharing; service engineering

I. INTRODUCTION

The recent worldwide economic depression has necessitated further streamlining of restaurant businesses to withstand severe price competition. Expanding restaurant chain stores is one means to increase sales and reduce business costs by providing similar dishes produced in a central kitchen. Japan has some restaurant chains of family restaurants, pub-style restaurants and sushi restaurants. However, unlike fast-food chains, expansion of full-service restaurants (dinner restaurants) is difficult because they require many human skills such as highly skilled cooks, complicated process management of customer orders, and good communication with customers. Hospitality is important in restaurant businesses for customer satisfaction.

Although restaurant businesses face some intrinsic problems that differ from those of manufacturing industries, science and technology are expected to raise service productivity further and to enhance customer satisfaction by supporting actual businesses. The Japanese Ministry of

Economy, Trade and Industry in 2007, for instance, started a national project to promote service engineering to improve service sector productivity [1]. Supported by the project, the authors initiated studies of problems in retail and restaurant businesses. We specifically elucidated service process management in actual restaurants based on objective data gathered by information systems and measuring apparatus.

Information technologies have played important roles in restaurant businesses in some aspects such as in inventory control [2] and supply chain management [3]. These motivations coincide with those of retail or manufacturing businesses. In Japan, point-of-sale (POS) systems originally developed for retail and CVS (or Convenience store) businesses were introduced into restaurant businesses [4] in the 1980s. Unlike the usage of POS system in CVS, that of restaurants is usually for management of customer orders. Although this system was innovative for restaurants, other problems remained. In such a POS system, orders are recorded by table and not by dish. When the restaurant is busy, many orders are accepted one after another. In such circumstances, kitchen staff make decisions about sequencing of preparing considering the type of dish. It is difficult for cooking staff to grasp the table number and to know how long each table has waited for service. Moreover, service staff must monitor the situation of their customers and the kitchen. Delayed delivery lowers customer satisfaction and sometimes elicits complaints. Because restaurant situations vary from time to time, process management heavily depends on staff skills and memory.

To resolve those problems, we propose a novel system by expanding conventional POS systems in which staff can share order information in real-time using an information system. Information sharing is a crucial factor of process management in a restaurant.

The next section overviews existing methods for process management in restaurants and elucidates problems underlying conventional methods.

II. PROBLEMS OF PROCESS MANAGEMENT IN RESTAURANTS

In simple terms, restaurant service processes comprise order receiving, preparing, and serving. An important task is to serve customers at the right place at the right time. However, that task is not easy when the restaurant becomes busy: many orders come from different tables at different times. Restaurant work is demanding, requiring a good memory and fast judgment. In general, in restaurants, three methods for process management have been used: verbal confirmation, order sheet, and a POS system.

A. Verbal confirmation

This method is the oldest but still used in small restaurants for service process management. When a server receives an order from a customer, it is memorized and related to the kitchen staff verbally. The kitchen staff prepares the order, sometimes in random order. The server is called and the order is given to the server. The server recalls the table and the customer who ordered the dish and serves the dish to the customer. The kitchen staff workers often ignore table information.

This method deeply depends on human skills; it is easy to make a mistake when many orders come almost simultaneously. Consequently, the restaurant scale depends on the capacity of human memory. This is one reason why the scale of many restaurants remains small.

B. Order sheet

By this method, when servers receive orders, they write them to an order sheet with a table number and hand the sheet to kitchen staff. Then, the kitchen staff prepares dishes according to the sheets. When they finish preparing them, they hand them to a server with the sheet. Then the server takes the dishes to the table written on the sheet.

In this method, the kitchen staff cannot distinguish the sequence of dishes because the order-entry time is not written in the sheet. The customers might become irate if a later-ordered dish is served to other tables earlier than their own. Moreover, the order sheets are often lost in an actual restaurant. When the sheet is lost, the staff cannot prepare or serve dishes. This often results in customer complaints.

C. POS system

Fig. 1 presents the structure of POS systems widely used in restaurants in Japan today. Originally, POS systems in restaurants were developed to obviate the handing-over processes of customer orders from service staff to kitchen staff. The POS system records and manages order information by table. When the server receives an order, they input information to a small order-entry device, from which it is transmitted to POS terminal. Then the POS

terminal sends the information to the kitchen; each order is printed by table on a printer in the kitchen. The kitchen staff members check some sheets and start preparations. After finishing, they pass the dishes with sheets to a server. However, in actual restaurants, the kitchen staff workers often calculate the number of identical dishes irrespective of tables and prepare them collectively to boost efficiency. The preparation processes still depend on their experience.

In this way, conventional POS systems present problems of information presentation. Moreover, they cannot support temporal management of preparing and servicing processes.

For the reasons explained above, we developed a new system by extending restaurant POS systems. It particularly addresses real-time process management of services. The system concept and functions are presented next.

III. PROCESS MANAGEMENT SYSTEM (PMS) FOR RESTAURANTS

This section introduces functions of a new system (PMS) developed for full-service restaurants in Japan.

Of course, an important function of POS in restaurants is registration of orders to avoid accounting mistakes. As Stain [4] pointed out, POS in restaurants is intended to improve checkout functions to avoid mistakes. In addition to those basic functions, PMS especially targets the temporal aspects of process management and information sharing among staff. Those functions are summarized as follows.

1. Order checking function by dish at each kitchen

When the PMS terminal receives order information from service staff, the PMS records the receipt time of each order and recalculates the total number of dishes. The PMS system transmits to the order checking system (presented in Fig. 2) placed in each kitchen. Figure 3 portrays a sample of the display of order checking system at certain kitchen. For instance, this kitchen takes charge of coffee, cakes, and light meals. The vertical line shows kinds of dishes. The number of orders is shown at the left side of each dish in this display. When kitchen staff workers start preparing them, they can determine the number of each dish ordered. This function can improve productivity because of overlapped units in the preparing process.

2. Elapsed time display and delay warning functions

The PMS checks the order-received times of all orders and calculates the elapsed time by dish until the cooking staff finish preparing them. The horizontal axis of Fig. 3 shows the passage of time by table. Squares move to the right every 2 min and their color changes every 10 min (blue–yellow–red). When the elapsed time exceeds 20 min, an alarm sounds to notify staff of the delay. Kitchen workers usually check and grasp conditions of the process. They judge the priority of orders and prepare them appropriately.

3. Checking functions of all kitchen positions' status

The order checking system can refer to statuses of other positions. A Chief or Manager who grasps the delay of a

certain position can order other staff to help the busy position. Using this function, they can manage the load of each position, thereby reducing labor costs to improve productivity by mobilization of staff positions. Especially, this function is available for large-scale restaurants or hotels that have plural kitchens [5]. In such cases, a chef in a certain kitchen cannot grasp the situation of other kitchens.

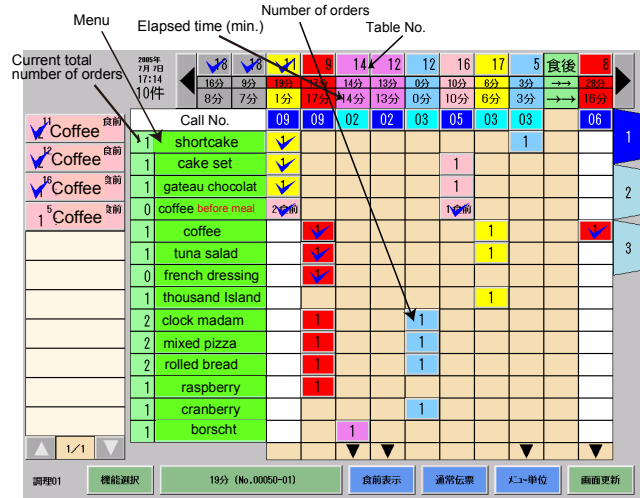


Figure 3. Layout for order-checking system

4. Search function and reissue for lost order sheet

Unlike conventional POS systems in restaurants, PMS does not print order sheets when it receives orders. Instead, on the order checking system, order sheets are printed automatically when a kitchen worker touches the dish symbol on a screen after preparation. The order sheets are used when the server delivers dishes to appropriate tables.

A server who loses an order sheet can search for information on the order checking system easily and reprint the order sheet. Using this function, staff need not search all tables to find customers who placed the order. In other words, PMS is more robust than conventional POS systems in preventing human error.

IV. INTRODUCTION OF PMS TO A RESTAURANT

To test the PMS system effectiveness, we introduced it in a restaurant, *Sanjo-Honten* (Ganko Food Service Co. Ltd., Osaka, Japan), during April 6–13, 2009. This restaurant is large, including four floors (each floor has 100 sheets), and three kitchens (a sushi counter on the first floor, an open kitchen on the second floor, and a backyard kitchen on the fourth floor). In this restaurant, it was particularly difficult for staff to grasp the situations of other floors because they were unable to see all positions simultaneously.

Before introducing the system, staff received PMS system operational training for a day to operate it smoothly.

In this experiment, serving times of dishes to customer tables, and customer claims per day were measured to assess

quantitative improvements. The PMS system comprises a PMS terminal, 10 order entry devices for service staff, and 8 order checking systems (displays) and printers in kitchens. Before introducing the new system, we measured the same parameters using the previous system—a conventional POS system—during March 23–29. After the experiment, 20 staff (6 system engineers, 8 cooking staff, and 6 service staff) were interviewed after the experiment to assess the qualitative efficiency for store operations and to check the PMS system load handling.

Results of PMS introduction

This section reports results of PMS system introduction. First, the average serving time from taking the order to service at a table was improved from 20 min to 16 min. Moreover, customer complaints about delayed service per day dropped from 15 to 3, demonstrating improvements not only in efficiency but also in customer satisfaction.

Table 1 presents samples of staff interviews in reference to four PMS functions.

TABLE I. Sample of comments in interview to staff

Function	Comment
Order checking function	The PMS displays all orders. Therefore, I did not need to calculate the dishes. (Kitchen staff) As operations improved using the system, productivity increased and the load was reduced (Kitchen staff)
Elapsed Time display function	It was easy to grasp the transition of each order, preventing preparation sequence mistakes. (Kitchen staff) The print function using the touch panel was awkward to use. Double-clicking sometimes occurred. (Kitchen staff)
Checking function of other positions	We were able to shift kitchen staff to busy positions for smooth operations. (Chef) We were able to detect crowds at each floor, and control employee shifts and entry of customers. (Manager)
Reissue function	It was easy to find the table—when losing order sheets—by using the search and reissue function. (Service staff) Because preparation and serving times were shortened, customer claims decreased drastically. (Service staff)

Along with positive effects of PMS, this experiment revealed some problems, especially in hardware systems. For example, kitchen workers were sometimes unaware of alarm sounds because of kitchen noise. The sound volume must be increased. Also, kitchen staff reported difficulty seeing the order checking system display because of light reflection. The monitor orientation must be improved.

V. DISCUSSION

This section presents discussion of the reasons why PMS improved the restaurant operation.

Originally, CVS industries such as convenience stores or pharmacies sought optimization of merchandizing systems by introducing POS systems. The POS data (sales) were used mainly outside business hours and outside of CVS

stores themselves. The CVS industry value chain is shown in Fig. 4. The value chain loop circles the CVS shop, CVS headquarters, vendors, and distributors.

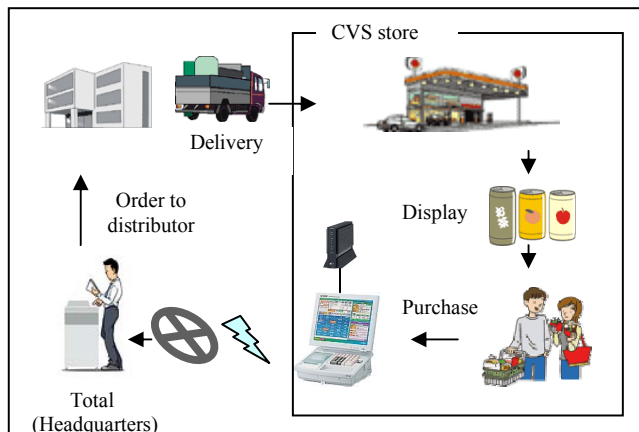


Figure 4. Value chain of CVS

In restaurant industries, POS systems were used for the same purpose: POS data were used after hours to assess sales. However, in restaurants, order information (POS data) is also important during business hours in restaurants.

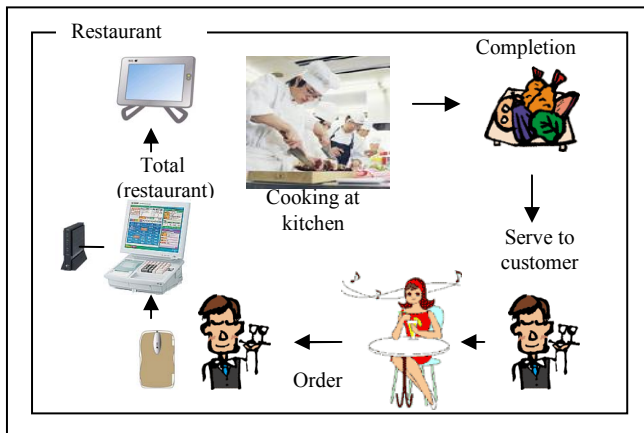


Figure 5. Restaurant value chain

Figure 5 presents the restaurant industry value chain. This circuit loops inside of a restaurant: customers, servers, kitchen staff, servers, and customers. However, POS data in conventional POS systems were not looped in this circle: information was used only for sales management after hours.

The PMS applies this information loop to the restaurant value chain. As introduced in section III, PMS has an order checking function, by which the kitchen staff can know all order information in real time, enabling them to streamline preparation processes and to reduce preparation times. Reducing customers' waiting times is an important factor related to customer satisfaction [6–7].

As Davis et al. described [8], if customers receive appropriate action concerning service, then they can wait a long time to serve a dish without frustrating the customer.

Customer satisfaction comprises many factors: waiting time, taste of food, and communication with staff. As shown, PMS enhances both efficiency and customer satisfaction.

VI. CONCLUSION

This paper introduces a new process management system (PMS) which augments POS systems for restaurants. In restaurant businesses, real-time management of customer orders is crucial. To that end, PMS provides an order checking function by dish and a display function of elapsed time. Moreover, the checking function of other positions' status helps managers to optimize staff assignments. By introducing this system to an actual restaurant, we confirmed that the average serving time was improved and that complaints decreased markedly. In fact, PMS contributed to both efficiency and customer satisfaction; they do not conflict but exist in the same context.

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